

Syllabus:- Straight Line

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The point P ($\alpha, \alpha + 1$) will lie inside the triangle with vertices A(0,3), B(-2,0) and C(6,1) if
 (1) $\alpha = -1$ (2) $\alpha = 2$ (3) $-\frac{3}{4} < \alpha < \frac{2}{7}$ (4) $-\frac{6}{7} < \alpha < \frac{3}{2}$
- A line passing through point (2, 2) encloses an area of magnitude λ with the axes (not in first quadrant). The intercepts made by the line on the axes are roots of equation
 (1) $x^2 - \lambda x + 2\lambda = 0$ (2) $x^2 + \lambda x - 2\lambda = 0$ (3) $x^2 + \lambda x + 2\lambda = 0$ (4) $x^2 + \lambda x + \lambda = 0$
- The new position of the point (1, 2) under rotation through an angle of 90° about the origin in anticlockwise direction is
 (1) (-2, 1) (2) (2, -1) (3) (1, -2) (4) (-1, 2)
- If the three distinct points $(t_i, 2at_i + at_i^3)$ for $i = 1, 2, 3$ are collinear then the sum of the abscissae of the points is
 (1) -1 (2) 0 (3) 1 (4) 3
- Let (a, b) and (c, d) are two points in a plane. Any point on the line joining these points has coordinates ($k \in \mathbb{R}$)
 (1) $(a + kc, b + kd)$ (2) $(ka + c, kb + d)$
 (3) $((1-k)a + kc, (1-k)b + kd)$ (4) $(a + (1-k)c, b + (1-k)d)$
- The points $(-a, -b)$, (0, 0), (a, b) and (a^2, ab) are
 (1) collinear (2) vertices of a parallelogram
 (3) vertices of rectangle (4) none of these
- The orthocentre of triangle with vertices (0, 0), (3, 4) and (4, 0) is
 (1) (3, 4) (2) (3, 12) (3) (3, 3/4) (4) (3, 9)
- The straight lines $x + y = 0$, $3x + y = 4$ and $x + 3y = 4$ form a triangle which is
 (1) isosceles (2) equilateral (3) right angled (4) none of these
- The intercept cut off from y-axis is twice that from x-axis by the line and line passes through (1,2) then its equation is
 (1) $2x + y = 4$ (2) $2x + y + 4 = 0$ (3) $2x - y = 4$ (4) $2x - y + 4 = 0$
- A line 'L' is drawn from (4,3) to meet the lines $L_1 : 3x + 4y + 5 = 0$ and $L_2 : 3x + 4y + 15 = 0$ at points A and B respectively. From 'A' a line, perpendicular to L is drawn meeting the line L_2 at A_1 . Similarly, from point 'B' a line, perpendicular to L is drawn meeting the line L_1 at B_1 . Thus parallelogram AA_1B_1 is formed. Least value of area of parallelogram AA_1B_1 is....
 (1) 4 (2) 2 (3) 6 (4) 8

11. If $x + y = 0$, $4x + 5y = 0$, $2x + y = 8$ are the sides of triangle ABC, then the line $x - 2y = 0$ passes through
 (1) incentre of $\triangle ABC$ (2) orthocentre of $\triangle ABC$
 (3) circumcentre of $\triangle ABC$ (4) centroid of $\triangle ABC$
12. Two vertices of a triangle are $(3, -2)$ and $(-2, 3)$, and its orthocentre is $(-6, 1)$. Then the third vertex of this triangle can NOT lie on the line :
 (1) $6x + y = 0$ (2) $4x + y = 2$ (3) $5x + y = 2$ (4) $3x + y = 3$
13. The sides of $\triangle ABC$ lie along lines $3x + 4y = 0$, $4x + 3y = 0$ and $x = 3$. Let (h, k) be the centre of the circle inscribed in $\triangle ABC$. The value of $h + k$ is
 (1) 0 (2) $1/4$ (3) $-1/4$ (4) $1/2$
14. Equation of straight line $ax + by + c = 0$, where $3a + 4b + c = 0$, which is at maximum distance from $(1, -2)$, is
 (1) $3x + y - 17 = 0$ (2) $4x + 3y - 24 = 0$
 (3) $3x + 4y - 25 = 0$ (4) $x + 3y - 15 = 0$
15. If the point $(p, 5)$ lies on the line parallel to the y-axis and passing through the intersection of the lines $2(a^2 + 1)x + by + 4(a^3 + a) = 0$ and $(a^2 + 1)x - 3by + 2(a^3 + a) = 0$ then p is equal to :
 (1) $3a$ (2) $-2a$ (3) $-3a$ (4) $2a$
16. Chord of the curve $4x^2 + y^2 - x + 4y = 0$ which subtend a right angle at the origin pass through a fixed point whose co-ordinate are
 (1) $\left(\frac{1}{5}, \frac{-4}{5}\right)$ (2) $\left(\frac{-1}{5}, \frac{4}{5}\right)$ (3) $\left(\frac{1}{5}, \frac{4}{5}\right)$ (4) $\left(\frac{-1}{5}, \frac{-4}{5}\right)$
17. If (x_1, y_1) be the centroid of the triangle formed by a line L and coordinate axes then equation of line L is
 (1) $\frac{3x}{x_1} + \frac{3y}{y_1} = 1$ (2) $xx_1 + yy_1 = 3$ (3) $\frac{x}{3x_1} + \frac{y}{3y_1} = 1$ (4) $\frac{x}{x_1} + \frac{y}{y_1} = 1$
18. A ray of light coming from the point $(1, 2)$ is reflected at a point A on the x-axis and then passes through the point $(5, 3)$. The coordinates of the point A is
 (1) $(5/13, 0)$ (2) $(-7, 0)$ (3) $(13/5, 0)$ (4) none of these
19. Let $A \equiv (0, 0)$, $B \equiv (4, 4)$ and $C \equiv (8, 2)$ be the vertices of a $\triangle ABC$. If straight lines through the points B and C meet at point P such that area of $\triangle PBC = \frac{1}{3}$ area of $\triangle ABC$, then locus of P can be
 (1) $x - 4y + 4 = 0$ (2) $x + 2y = 32$ (3) $x + 2y = 8$ (4) $x - y - 2 = 0$
20. Consider a $\triangle ABC$ such that its incircle touches the base BC at $D(3, -3)$. If the equation of angle bisectors of angle A and B of the triangle are $x + y = 1$ and $2x - y = 5$ respectively, then equation of BC is
 (1) $2x + y - 3 = 0$ (2) $x - 2y - 9 = 0$ (3) $2x - 3y - 15 = 0$ (4) $x + y = 0$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Find the number of the values of k for which the lines $kx + 2y + 2 = 0$, $2x + ky + 3 = 0$, $3x + 3y + k = 0$ are concurrent
22. Find the distance of point $(4, 5)$ from the straight line joining the points $(1, 2)$ and $(-2, 3)$ measured parallel to the line $x + y + 1 = 0$
23. If ratio of slopes of lines in $kx^2 + 4xy + 3y^2 = 0$ is 2, then find the value of ' k '
24. If the equation $2x^2 + 2hxy + 6y^2 - 4x + 5y - 6 = 0$ represents a pair of straight lines then find the length of intercept on the x -axis cut by the lines
25. Let H be the orthocenter of $\triangle ABC$ where $A \equiv (0, 0)$, $B \equiv (8, 0)$ and $C \equiv (4, 4\sqrt{3})$. If $G_1, G_2, G_3, \dots$ are centroids of $\triangle AHB, \triangle AG_1B, \triangle AG_2B, \dots$ respectively then find value of $\lim_{n \rightarrow \infty} \sum_{i=1}^n \text{Ar } \triangle AG_iB$
- [Note : $\text{Ar}(\triangle AG_iB)$ denotes area of $\triangle AG_iB$]

Answer Key_DPP-07

1.	(4)	2.	(2)	3.	(1)	4.	(2)	5.	(3)	6.	(1)	7.	(3)
8.	(1)	9.	(1)	10.	(4)	11.	(2)	12.	(3)	13.	(1)	14.	(4)
15.	(2)	16.	(2)	17.	(3)	18.	(3)	19.	(3)	20.	(2)	21.	(2)
22.	(8.48)	23.	(1.18 or 1.19)	24.	(4)	25.	(4.61 or 4.62)						